

The Five Assumptions of Linear Regression

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Multicollinearity ([Note ML-026](#)); Q-Q plot ([Note ML-029](#)).

1. Overview

Key point: Linear regression is reliable when five conditions hold: a linear relationship, no multicollinearity, normal residuals, constant spread of residuals, and no pattern in the residuals over the rows.

Linear regression always produces a line or hyperplane, whatever the data looks like. Whether that line can be trusted depends on a few conditions, called the **assumptions** (G-220) of linear regression. Like the rules of a game, they do not stop anyone from playing, but break them and the score means little.

Textbooks list them in slightly different ways; these five are the core ones (ISL §3.3.3; Kutner Ch. 3 and §7.6):

	Assumption	Checked with
1	Linear relationship between each feature and the output	scatter plots
2	No multicollinearity between the features	VIF, correlation heatmap
3	Residuals are normally distributed	histogram, Q-Q plot
4	Homoscedasticity: residuals have the same spread everywhere	residuals against predictions
5	No autocorrelation of the residuals	residuals in row order

Three words first. A **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) is the output we predict.

Recall from [the equation of multiple linear regression](#) the symbols used below. The model predicts:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots$$

Here $\hat{y}$ (“y hat”) is the predicted target, y the actual target, $x_1, x_2, \dots$ the features and each β a coefficient (a learned number). The first two assumptions concern the features; the last

three concern the **residuals** (G-705), the errors $y_i - \hat{y}_i$ on each observation. Figure 1 draws them for the model below (each dot is one observation: its actual target against the model's prediction): each stick is the gap between an observation's actual target and the model's prediction.

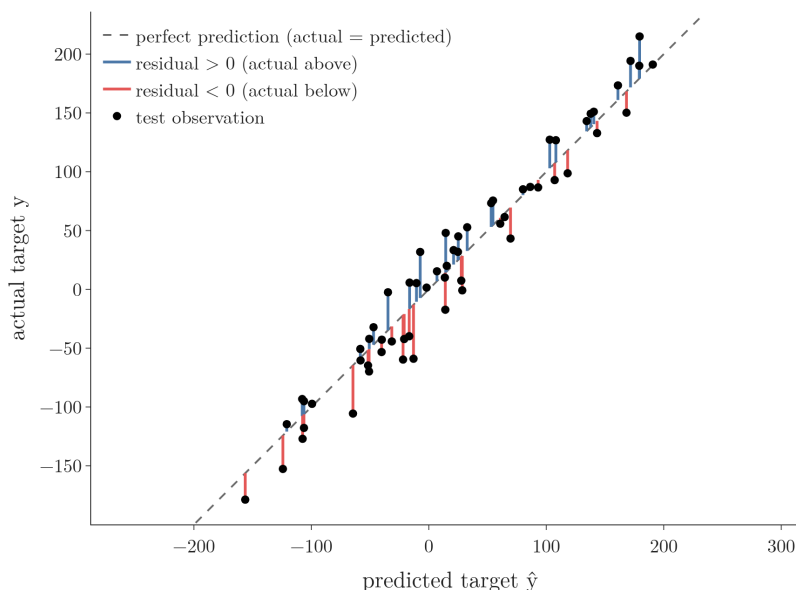


Figure 1: The 60 test observations: actual target against predicted target. Each stick is one residual, the vertical gap to the dashed diagonal of perfect predictions (blue: actual above, red: actual below). Assumptions 3 to 5 are about these sticks.

The example data has 200 observations, three features and one target. A linear regression trained on 70% of it scores $R^2 = 0.96$ on the other 30% (60 observations); each check below uses those 60 test residuals.

```
Python: The model whose assumptions we check.
df = pd.read_csv("data/data.csv")
X, y = df.iloc[:, :3].to_numpy(), df["target"].to_numpy()
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)
model = LinearRegression().fit(X_train, y_train)
residual = y_test - model.predict(X_test)
```

2. Assumption 1: a linear relationship

Key point: Each feature should relate to the output along a straight line, rising or falling. A curved relationship needs a different model.

Linear regression fits straight lines, so the relationship between each feature and the output should be roughly linear. Both directions are fine:

- **positive:** the output rises as the feature rises;
- **negative:** the output falls as the feature rises.

A curved relationship (for example, the output growing with the square of the feature) breaks this assumption, because a straight line then misses the pattern systematically.

The check is a scatter plot of the output against each feature (Figure 2).

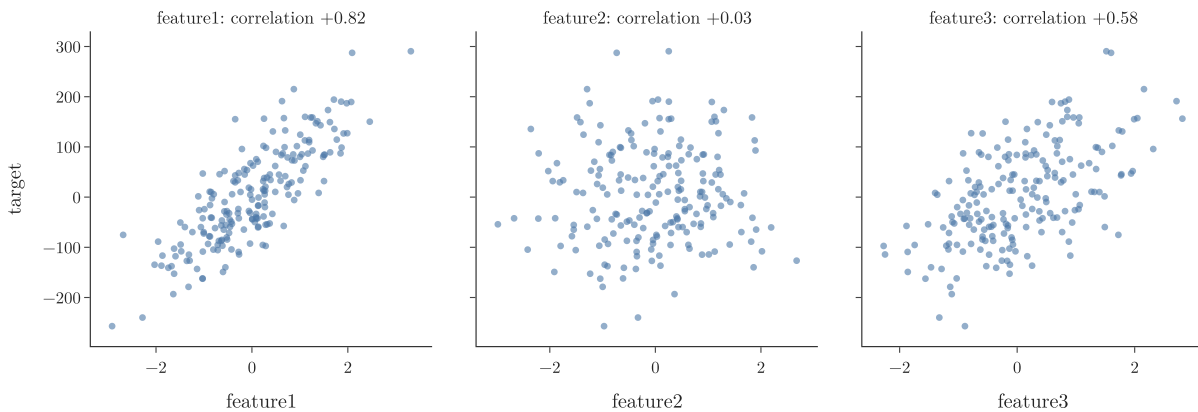


Figure 2: Each feature against the target

- **feature1:** a clear rising line (**correlation** (G-490) 0.82).
- **feature3:** also rising, with more scatter (0.58).
- **feature2:** no visible relationship at all (0.03). Its fitted coefficient is only -0.28 , against 72.7 and 53.3 for the other two: the model has learned to almost ignore it.

Extra: A feature with no relationship, like feature2, does not break the assumption so much as add nothing; the model gives it a near-zero weight. When a feature has a curved relationship, a simple approach is to add non-linear transformations of that feature, such as $\log x$, $\sqrt{x}$ or x^2 , to the model (ISL §3.3.3). Adding x^2 is polynomial regression ([adding powers as new features](#)).

3. Assumption 2: no multicollinearity

Key point: The features should not be strongly related to each other. If they are, the model cannot tell which one deserves the credit, and the coefficients become unreliable.

Multicollinearity (G-1273) means one feature can be (largely) predicted from the others. Multicollinearity was met as [the dummy variable trap](#) in one-hot encoding.

3.1 Why it is a problem

Key point: Each coefficient measures the effect of its feature with the others held fixed. Related features cannot be held fixed separately.

A coefficient β_1 is the change in the output when x_1 rises by 1 *and all other features stay the same*. If x_1 and x_3 always move together, “change x_1 and keep x_3 fixed” never happens in the data. The model then cannot separate their effects.

An analogy: two scientists finish a project together. If one is a physicist and the other a chemist, it is easy to say who contributed what. If both have identical skills, it is impossible. With related features the model faces the second case, so the coefficients become unreliable: a small change in the data can move them a long way (ISL §3.3.3). The predictions, on the other hand, usually stay good (Kutner §7.6).

Figure 3 tests this on our data. We refit the model on 20 **bootstrap** (G-320) resamples of the training data: each resample draws 140 observations from the 140 with replacement, so it differs a little from the original. With the Note’s own features (left), feature1’s coefficient stays between 69.9 and 78.0. Then we add a near-copy of feature1, feature1 plus a little noise (VIF 429, Section 3.2). Now feature1’s coefficient swings from 21.5 to 102.5 and the copy’s swings the opposite way, sometimes below 0. Their sum stays between 69.7 and 78.0: the model knows the pair’s joint effect, but not how to split the credit.

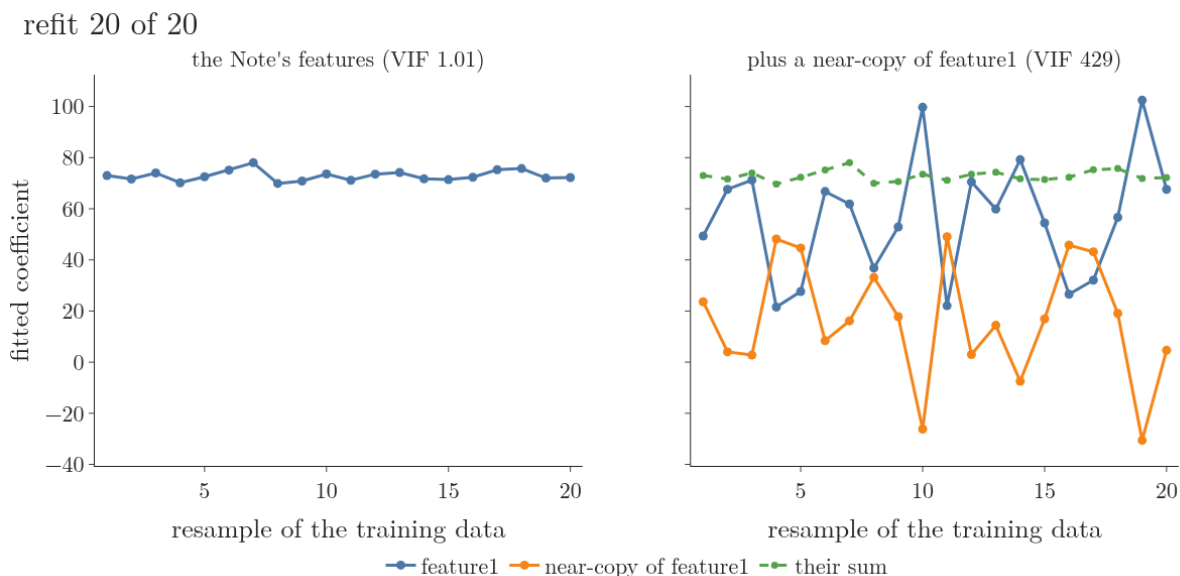


Figure 3: Refitting on 20 bootstrap resamples of the training data. Left: the Note’s features; feature1’s coefficient barely moves. Right: with a near-copy of feature1 added, the two coefficients swing wildly in opposite directions, while their sum (dashed green) stays steady.

3.2 Checking it

Key point: A VIF near 1 means no multicollinearity; above 5 is a problem. All three features here have VIF 1.01.

The standard check is the **variance inflation factor (VIF)** (G-2075). For each feature, a regression predicts that feature from all the other features; if they predict it well (high R^2), the feature is redundant.

In words: the VIF of a feature is one divided by the part of it the other features cannot explain.

$$\text{VIF of feature } j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 of predicting feature j from the others. With numbers: if the others explain 80% of a feature ($R_j^2 = 0.8$), its VIF is the usual danger line:

$$\frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$$

If they explain none of it, the VIF is 1.

Python: VIF with statsmodels.

```
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)
from statsmodels.tools import add_constant

Xc = add_constant(X_train)      # add an intercept column
[variance_inflation_factor(Xc, i) for i in range(1, 4)]
# [1.011, 1.010, 1.014]
```

A quicker, rougher check is a **heatmap** (G-886) of the correlations between the features (Figure 4, left): a table of correlations in which each cell pairs the feature of its row with the feature of its column, coloured darker the closer the value is to 1. The diagonal is 1 because each feature matches itself. Here every correlation between different features is at most 0.06: no multicollinearity.

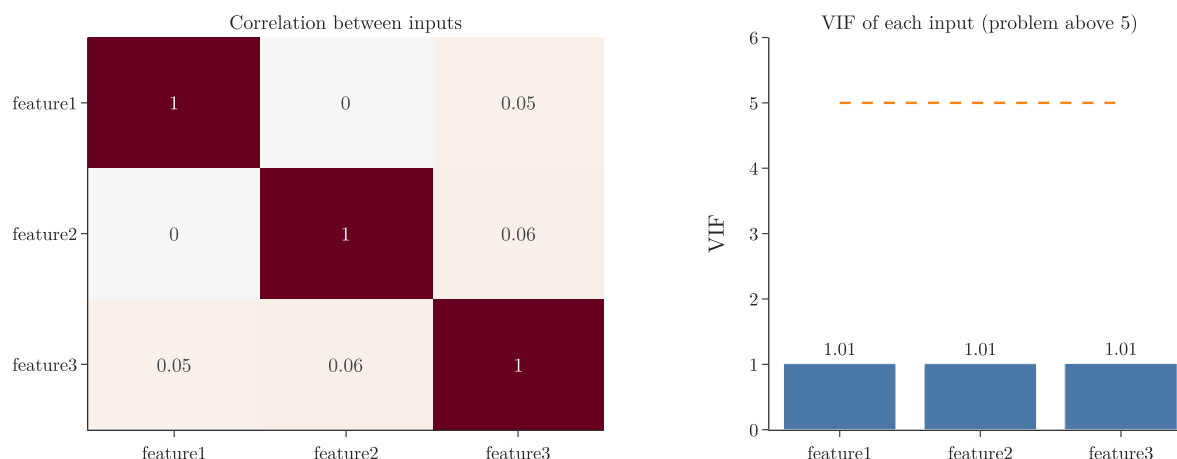


Figure 4: Correlations between the features, and their VIF

Extra: If VIF is high, two simple fixes are to drop one of the related features, or to combine them into one feature, such as their average after scaling (ISL §3.3.3). Ridge regression ([the idea: penalise large coefficients](#)) was also designed for related features (Hoerl and Kennard).

4. The residual plot

Key point: A residual plot puts every residual above or below a flat zero line. Random scatter around zero means the model fits; any shape means an assumption is broken.

The last three assumptions are all checked on one kind of picture, so we build it by hand first. Take four points, (1, 2), (2, 3), (3, 7) and (4, 8). Their least-squares line is $\hat{y} = 2.2x - 0.5$.

1. **Compute each residual**, actual minus predicted:

x	actual y	predicted $\hat{y} = 2.2x - 0.5$	residual
1	2	1.7	+0.3
2	3	3.9	-0.9
3	7	6.1	+0.9
4	8	8.3	-0.3

2. **Draw a flat line at 0.** It stands for the fitted line.
3. **Plot each residual** at its x : above the zero line when the point lies above the fitted line, below when it lies below.

Figure 5 does these steps one point at a time. The result is the **residual plot** (G-2207): the fitted line laid flat, with only the errors left to look at.

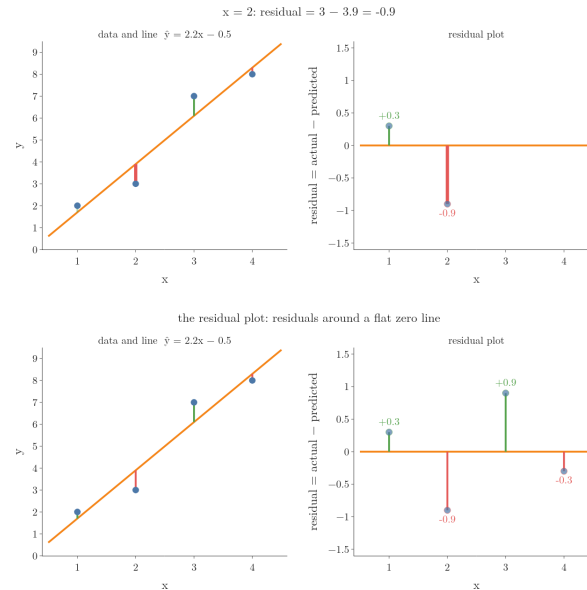


Figure 5: Building a residual plot from four points. Left: the points, the least-squares line and each residual as a stick (green above the line, red below). Right: the same sticks moved onto a flat zero line. Idea after Khan Academy, “Residual plots”.

How to read a residual plot:

- **Points scattered evenly above and below 0, with no shape:** the line describes the data well.
- **A curve,** for example the residuals sit high on the left, low in the middle and high again on the right (above the dashed zero line, then below it, then above again): the relationship is not a straight line. Assumption 1 is broken, and a non-linear model such as [polynomial regression](#) fits better (Figure 7, left).
- **A funnel or a wave:** assumptions 4 and 5 below.

With several features there is no single x to put on the horizontal axis, so the predicted value $\hat{y}$ is used instead. The reading rules stay the same.

5. Assumption 3: normal residuals

Key point: The residuals should follow a bell-shaped (normal) distribution centred on 0: most errors small, few large, positive and negative alike.

When the model is right on average, its errors should scatter around 0: many small errors and few large ones, as often too high as too low. Such a bell-shaped spread is a **normal distribution** (G-1343).

Two checks, both on the residuals (Figure 6, top):

- **Histogram or density plot:** roughly a bell centred on 0. Here it is, apart from a small bump.

- **Q-Q plot** (G-1596; see [how a Q-Q plot is built](#)): each residual is plotted against the value a perfect bell would give, so the points should lie on the straight line. Here they do, with small wiggles.

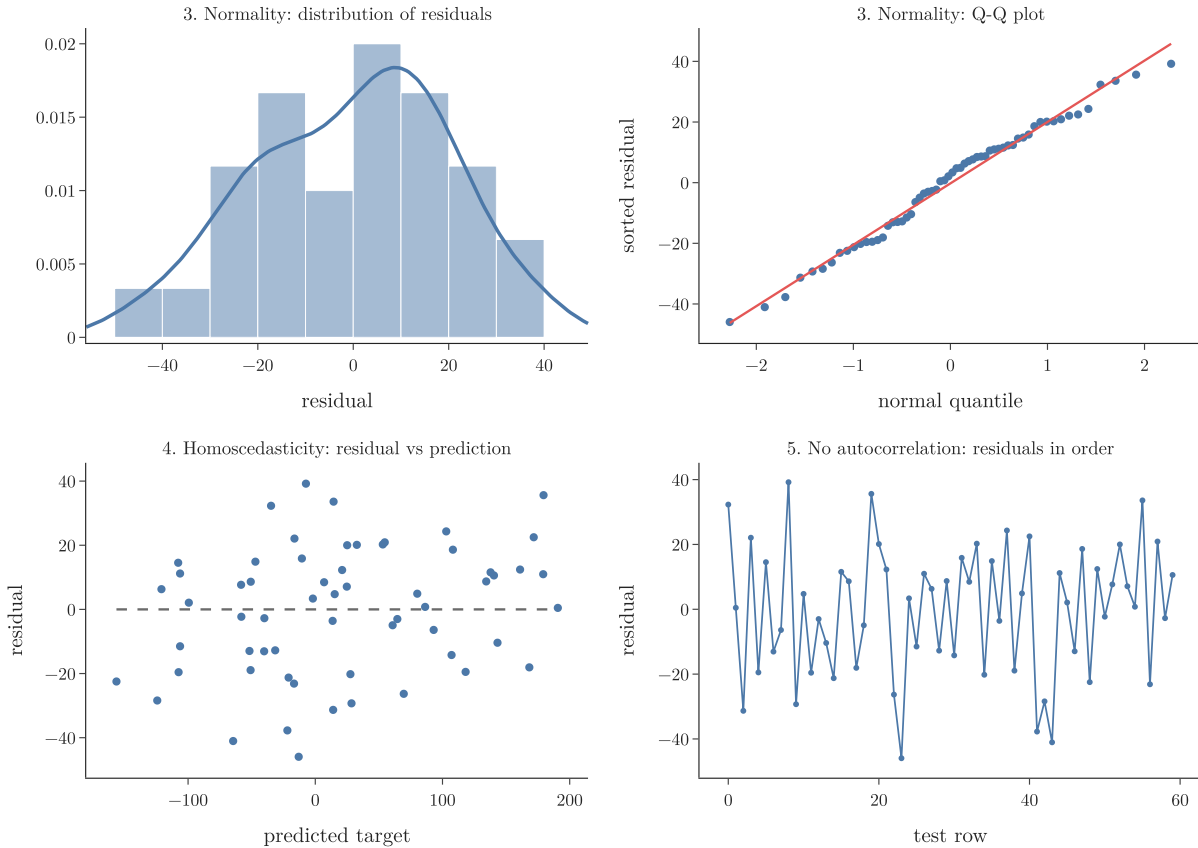


Figure 6: Residual checks for assumptions 3, 4 and 5

Extra: Formal tests exist as well. The **Shapiro-Wilk test** (G-1788) gives a **p-value** (G-1433; the chance of results at least this extreme if the data were normal, see [the definition](#)); above 0.05 means no evidence against normality. Here $p = 0.51$, and the skewness of the residuals is -0.23 , close to 0. With large samples these tests flag even small departures from normality, so look at the plots too (Ghasemi and Zahediasl 2012).

6. Assumption 4: homoscedasticity

Key point: The residuals should have the same spread for small and large predictions. A funnel shape (heteroscedasticity) breaks the assumption.

Homoscedasticity (G-902) means “same scatter”: the size of the errors does not depend on the size of the prediction. Its opposite, **heteroscedasticity** (G-889), is common (ISL §3.3.3): for example, a house-price model that is off by a few thousand on cheap houses and

by much more on expensive ones.

The check is a scatter plot of residuals against predicted values (Figure 6, bottom left). The plot should look like an even band around 0, with no shape. Here it does: the spread is roughly the same from the lowest to the highest prediction.

Figure 7 (middle) shows what heteroscedasticity looks like: a funnel that widens to the right.

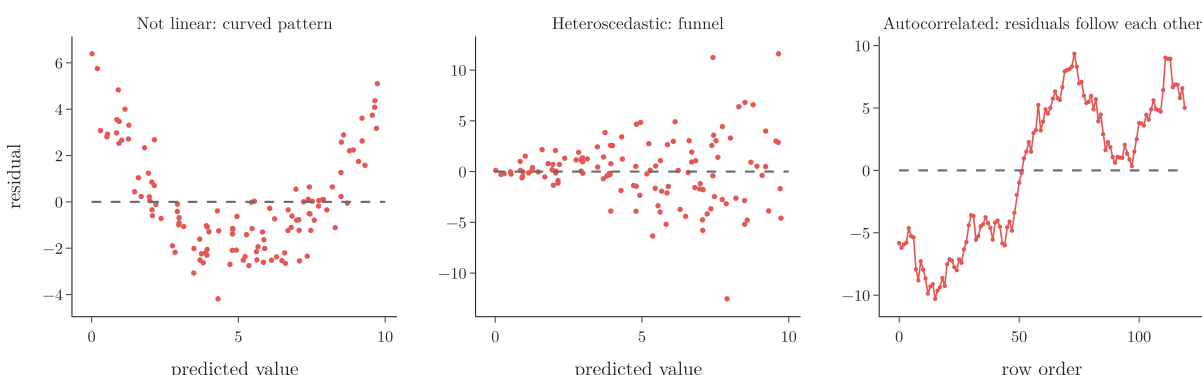


Figure 7: What failed assumptions look like (made-up data)

Extra: Heteroscedasticity does not make the coefficients wrong on average, but it makes the model's uncertainty estimates wrong: the confidence intervals and p-values that statistics packages report (Wooldridge §8.1). One fix is to take the log or square root of the target. The log shrinks large values more than small ones, and can turn a funnel into an even band (ISL §3.3.3).

7. Assumption 5: no autocorrelation of the residuals

Key point: One residual should not predict the next. Plotted in row order, the residuals should jump around randomly, not drift in waves.

Autocorrelation (G-230) means each residual is related to the one before it: if the model was too high on one observation, it is also too high on the next. Autocorrelation is common in **time series** (G-1975) data, measurements taken at points in time (ISL §3.3.3).

The check is to plot the residuals in row order (Figure 6, bottom right). They should jump up and down with no pattern. Here they do.

Figure 7 (right) shows positive autocorrelation: long runs above 0 followed by long runs below, like a slow wave. One cause is a missing feature that changes slowly from one observation to the next, such as a time trend: the model cannot see the trend, so the trend shows up in the residuals. Figure 8 shows this test (details in the Extra below): residuals of a model that misses a slow wave form the same wave; adding the wave as a feature leaves random jumps. Autocorrelation makes the model look more certain than it is: its confidence intervals come out too narrow (ISL §3.3.3).

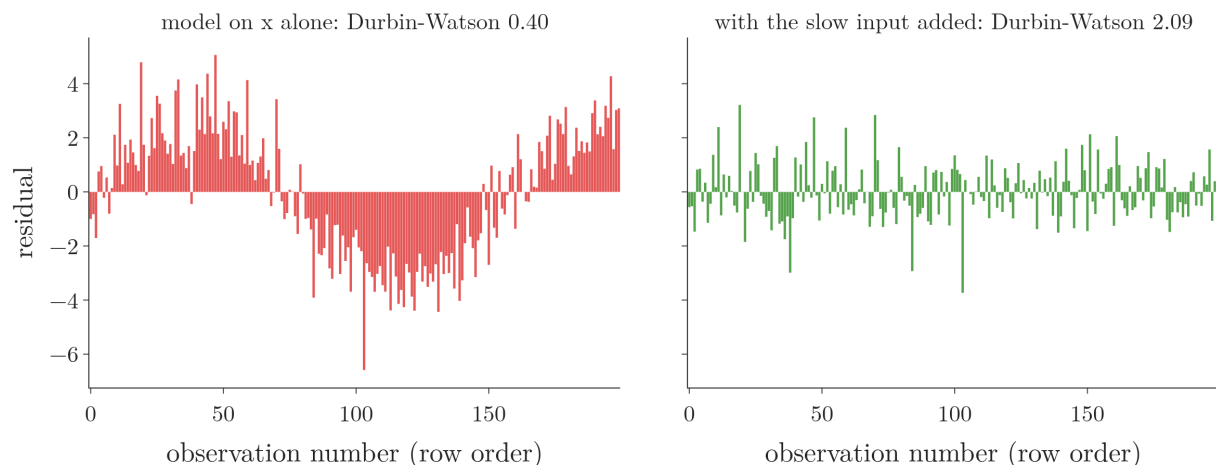


Figure 8: Residuals in row order for data built as $y = 2x + 3 \sin(t/25) + \text{noise}$. Left: a model on x alone leaves the slow wave in its residuals (Durbin-Watson 0.40). Right: with $\sin(t/25)$ added as a feature, the residuals jump randomly (Durbin-Watson 2.09).

Extra: The **Durbin-Watson statistic** (G-649) puts a number on autocorrelation: about 2 means no autocorrelation, towards 0 means positive autocorrelation, towards 4 negative (statsmodels docs). Here it is 2.31, close to 2: no sign of autocorrelation. To test the missing-trend cause, the notebook makes data from this rule, with t the observation number:

$$y = 2x + 3 \sin(t/25) + \text{noise}$$

Fitting on x alone gives Durbin-Watson 0.40; adding the slow wave $\sin(t/25)$ as a feature brings it back to 2.09. Figure 7 (left) also shows a residual plot when assumption 1 fails: a curve instead of a flat band.

8. Summary

Assumption	What it means	How to check	This data
1. Linearity	each feature relates to the output along a line	scatter plots	yes (feature2 unrelated)
2. No multicollinearity	features not related to each other	VIF (problem above 5), correlation heatmap	VIF 1.01
3. Normal residuals	errors form a bell around 0	histogram, Q-Q plot	yes
4. Homoscedasticity	errors have the same spread everywhere	residuals vs predictions	yes
5. No autocorrelation	errors do not follow each other	residuals in row order	yes

- Assumptions 1 and 2 are about the features; 3 to 5 are about the residuals.
- A failed assumption does not stop the model from running; it makes its coefficients or uncertainty estimates untrustworthy.
- Residual plots should look like random noise; any shape (curve, funnel, wave) points to a broken assumption.

9. Sources

Built from

- CampusX, “What are the main Assumptions of Linear Regression? | Top 5 Assumptions of Linear Regression”, YouTube, <https://www.youtube.com/watch?v=EmSNAtcHLm8>
- Khan Academy, “Residual plots”, YouTube, <https://www.youtube.com/watch?v=VamMrPZ-8fc> (building a residual plot by hand and reading it, section 4)

Other references

- **ISL**: James, Witten, Hastie, Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021. §3.3.3, Potential Problems.
- **Kutner**: Kutner, Nachtsheim, Neter, Li, *Applied Linear Statistical Models*, 5th ed., McGraw-Hill, 2005. Ch. 3 (diagnostics and remedial measures), §7.6 (multicollinearity and its effects).
- **Hoerl and Kennard**: A. E. Hoerl and R. W. Kennard, “Ridge Regression: Biased Estimation for Nonorthogonal Problems”, *Technometrics* 12(1), 55–67, 1970.
- **Ghasemi and Zahediasl**: A. Ghasemi and S. Zahediasl, “Normality Tests for Statistical Analysis: A Guide for Non-Statisticians”, *International Journal of Endocrinology and Metabolism* 10(2), 486–489, 2012.
- **Wooldridge**: J. M. Wooldridge, *Introductory Econometrics: A Modern Approach*, Cengage. Chapter 8, Heteroskedasticity (§8.1, consequences for OLS).
- **statsmodels docs**: statsmodels documentation, `statsmodels.stats.stattools.durbin_watson`.

10. Key terms

Term	Meaning
Feature	An input variable: one column of the data table
Observation	One record: one row of the data table
Target	The output we predict
Assumption (of a model)	A condition the data must meet for the model's results to be reliable
Residual (G-705)	The error on one data point: actual minus predicted value
Residual plot (G-2207)	The residuals plotted above and below a flat zero line, against a feature or the predicted value
Variance inflation factor (VIF)	$1/(1 - R_j^2)$: how well the other features predict feature j ; above 5 signals multicollinearity
Homoscedasticity	The residuals have the same spread for all predicted values
Heteroscedasticity	The spread of the residuals changes with the predicted value, often as a funnel
Autocorrelation	Each residual is related to the one before it in row order
Shapiro-Wilk test	A statistical test of whether data follows a normal distribution
Durbin-Watson statistic	A number from 0 to 4 measuring autocorrelation of residuals; about 2 means none
statsmodels	A Python library for statistical models and tests